Mario Ayala

Mario Antonio Ayala Valenzuela

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Research interests

Probability theory, and more precisely scaling limits of interacting particle systems: hydrodynamic limits, fluctuation fields and their SPDE limits, duality and orthogonal duality polynomials, Dirichlet forms and Mosco convergence, condensation. Effective dynamics and coarse-graining of stochastic systems, including the interpretation of multiplicative noise. Stochastic population models.

Positions

since 08/2022	Postdoctoral researcher, Chair for Analysis and Modelling, Technische Universität München, Germany. Host: Johannes Zimmer.
05/2021–07/2022	Postdoctoral researcher, BioSP, INRAE, Avignon, France. With Jérôme Coville and Raphaël Forien.

Education

2021	PhD in Applied Mathematics, Delft University of Technology, the Netherlands. Thesis: *Fluctuations for interacting particle systems with duality*, defended on 18 February 2021. Promotor Frank Redig, copromotor Gioia Carinci.
2016	MSc in Applied Mathematics, specialization in Financial Engineering, Delft University of Technology. Thesis: *Hydrodynamic limit for the symmetric inclusion process*.
2011	BSc in Mathematics, Facultad de Ciencias, Universidad de Colima, Mexico. Thesis: *Estimation of the basic reproductive number for HLB in the state of Colima, Mexico*.

Distinctions and scholarships

01/2022–12/2028	National Researcher, Level I, Mexican National System of Researchers (SNII).
PhD	CONACYT scholarship for PhD studies at Delft University of Technology.
MSc	CONACYT scholarship for MSc studies at Delft University of Technology.
2003	Bronze medal, 17th Mexican Mathematical Olympiad, Guanajuato.
2003	First place, national Mathematical Skills Contest, ITESO, Jalisco.

Publications

Published.

[1] M. Ayala and J. Zimmer. Hydrodynamic limits and non-equilibrium fluctuations for the symmetric inclusion process with long jumps. *Annales Henri Poincaré* **27**(7), 2685–2721 (2026).

[2] M. Ayala, J. Coville and R. Forien. A measure-valued stochastic model for vector-borne viruses. *ALEA Latin American Journal of Probability and Mathematical Statistics* **21**, 1145–1193 (2024).

[3] M. Ayala, G. Carinci and F. Redig. Condensation of SIP particles and sticky Brownian motion. *Journal of Statistical Physics* **183**(3), art. 40 (2021).

[4] M. Ayala, G. Carinci and F. Redig. Higher order fluctuation fields and orthogonal duality polynomials. *Electronic Journal of Probability* **26**, 1–35 (2021).

[5] M. Ayala, G. Carinci and F. Redig. Quantitative Boltzmann-Gibbs principles via orthogonal polynomial duality. *Journal of Statistical Physics* **171**(6), 980–999 (2018).

Submitted.

[6] M. Ayala, N. Dirr and J. Zimmer. Atomic limits of empirical measures of interacting particle systems with self-duality. `arXiv:2610.31319` (2026).

[7] M. Ayala and J. Zimmer. Computing extinction barriers in a quorum-sensing reaction network. `arXiv:2609.39946` (2026).

[8] M. Ayala and B. Vallejo Jiménez. Consumption-investment under alternative stochastic-integral conventions. `arXiv:2602.08527` (2026).

[9] M. Ayala and D. R. M. Renger. Hydrodynamic limits of collisions and fluxes in the exclusion process. `arXiv:2512.20412` (2025).

[10] M. Ayala, N. Dirr, G. A. Pavliotis and J. Zimmer. Reversibility, covariance and coarse-graining for Langevin dynamics: on the choice of multiplicative noise. `arXiv:2511.03347` (2025).

[11] M. Ayala. Mosco convergence of independent particles and applications to particle systems with self-duality. `arXiv:2406.19088` (2024).

[12] M. Ayala, J. Coville and S. Soubeyrand. Group dispersal modelling revisited. `arXiv:2405.08384` (2024).

In preparation.

[13] M. Ayala and F. Redig. Duality for SPDEs.

[14] M. Ayala and J. Zimmer. Cooperation through a shared environment: hydrodynamic limits and large deviations for cheater invasion.

[15] M. Ayala and J. Zimmer. Derivation of the symmetric exclusion process from a monitored Bose-Hubbard model.

[16] M. Ayala and D. R. M. Renger. Cumulants, SEP and criticality.

Theses and technical reports.

- M. Ayala. Fluctuations for interacting particle systems with duality. PhD thesis, Delft University of Technology (2021).
- M. Ayala. Hydrodynamic limit for the symmetric inclusion process. MSc thesis, Delft University of Technology (2016).
- Evaluation of *Aedes aegypti* control campaigns. Technical report for the Ministry of Health of the State of Colima, Mexico (2009).
- M. Ayala, C. L. Pawling, A. N. Smith, L. Gao, D. Hiebeler and B. R. Morin. An epidemiological approach to the dynamics of chytridiomycosis on a harlequin frog population. Technical report, Mathematical and Theoretical Biology Institute, Arizona State University (2006).

Talks

09/2026 — *Derivation of the symmetric exclusion process from a monitored Bose-Hubbard model*. MMSC 2026: Workshop on Mathematical Modeling and Scientific Computing, Technische Universität München, Germany.

08/2025 — *Del orden al azar: modelando la incertidumbre en economía y más allá*. Invited talk, Universidad de Colima, Mexico.

07/2025 *Hydrodynamic limits and non-equilibrium fluctuations for the symmetric inclusion process with long jumps*. Seminar, Universität Hamburg, Germany.

04/2025 *Hydrodynamic limits and non-equilibrium fluctuations for the symmetric inclusion process with long jumps*. Workshop "Fluctuating hydrodynamics", TUM, Raitenhaslach, Burghausen, Germany.

07/2024 *Scaling limits for the inclusion process with long jumps*. University of Modena and Reggio Emilia, Italy.

04/2023 *Condensation of SIP particles and sticky Brownian motion*. Workshop "In search of model structures for non-equilibrium systems", Münster, Germany.

03/2023 *Fluctuation fields and orthogonal self-dualities*. Seminário de Probabilidade e Mecânica Estatística, online.

11/2021 *A spatial stochastic model for a vector-borne virus population: some scaling limits*. Séminaire Mathématiques, Évolution, Biologie, Aix-Marseille Université, France.

08/2021 *A spatial stochastic model for a vector-borne virus population*. YEP XVII: Interacting particle systems, Eurandom, Eindhoven, the Netherlands.

07/2021 *Higher order fluctuation fields and orthogonal duality polynomials*. Encontro Nacional da SPM 2021, session on stochastic duality for Markov processes, online.

02/2020 *Higher order fluctuation fields and orthogonal duality polynomials*. Third Haifa Probability School, Workshop on Random Geometry and Stochastic Analysis, Technion, Haifa, Israel.

05/2019 *Condensation of SIP particles and sticky Brownian motion*. 27th Meeting of PhD students in Stochastics in the Netherlands.

05/2019 *Condensation of SIP particles and sticky Brownian motion*. NDNS+ PhD Days 2019, the Netherlands.

02/2018 *Sistemas de partículas que interactúan y aplicaciones a modelos de distribución de la riqueza*. Internal seminar, Facultad de Economía, Universidad de Colima, Mexico.

08/2006 *An epidemiological approach to the dynamics of chytridiomycosis in a harlequin frog population*. SIAM Conference on the Life Sciences, Raleigh, USA.

Posters.

07/2019 *Condensation of SIP particles and sticky Brownian motion*. LMS Research School "Random structures: from the discrete to the continuous", University of Bath, United Kingdom.

10/2006 *An epidemiological approach to the dynamics of chytridiomycosis in a harlequin frog population*. SACNAS National Conference, Tampa, USA.

Schools and conferences attended

02/2020 Third Haifa Probability School, Workshop on Random Geometry and Stochastic Analysis, Technion, Haifa, Israel.

07/2019 LMS Research School "Random structures: from the discrete to the continuous", University of Bath, United Kingdom.

07/2018 CIMPA School "Geometry and scaling of random structures", Universidad de Buenos Aires, Argentina.

04/2017	Pre-school “Stochastic dynamics out of equilibrium”, CIRM, Marseille, France.
09/2016	VIII Escuela de Probabilidad y Procesos Estocásticos, CIMAT, Guanajuato, Mexico.
09/2015	7th General AMaMeF and Swissquote Conference, Lausanne, Switzerland.
summer 2006	Mathematical and Theoretical Biology Institute (MTBI), Arizona State University, USA.

Teaching

WS 2026/27	Exercise coordinator, *Basic Mathematics I*, a course for engineering students, Technische Universität München. Approximately 300 students.
WS 2024/25	Exercise coordinator, *Functional Analysis*, Technische Universität München. Coordination of the tutors, design of problem sheets and exams; more than 240 students.
SS 2023, SS 2024	Exercise coordinator, *Calculus III: Integral and Differential Calculus*, Technische Universität München. Coordination of the tutors, design of problem sheets and exams; more than 580 students.
WS 2022/23, WS 2023/24	Tutor, *Functional Analysis*, Technische Universität München.
2019–2020	Teaching assistant, Delft University of Technology: *Stochastic Differential Equations* (spring 2020), *Linear Algebra* (winter 2019/20), *Analyse en differentiaalvergelijkingen* (autumn 2019).
08–12/2018	Lecturer, *Numerical Methods* (in Spanish), Universidad de Colima, Mexico.
since 2005	Trainer for the Mexican Mathematical Olympiad in Colima, Mexico, on and off: regularly in 2005–2010, and online since 2021.

Service

Referee	*The Annals of Probability*, *The Annals of Applied Probability*, *Electronic Journal of Probability*, *Probability and Mathematical Physics*, *Journal of Statistical Physics*, *Proceedings of the Royal Society A*, *Brazilian Journal of Probability and Statistics*, *Markov Processes and Related Fields*.
11/2022	Mentor of the TUM team “Quod erat demonstranTUM” at the International Student Team Competition in Mathematics (ISTCiM 2022), University of Silesia, Katowice, Poland: bronze medal and third prize.

Other professional experience

07–08/2017	Consultant, Dubai Community Development Authority and Dubai Airports Authority, United Arab Emirates.
03–07/2016	Intern, Dutch State Treasury Agency, Ministry of Finance, the Netherlands. Analysis of Dutch inflation-linked bonds.
2007–2014	Administrative support and operations, Ministry of Health of the State of Colima, Mexico.

Languages and computing

Languages	Spanish (native), English (fluent), German (A2).
Computing	Python, R, MATLAB, LaTeX.

Last updated: October 2026.